

Neural Networks

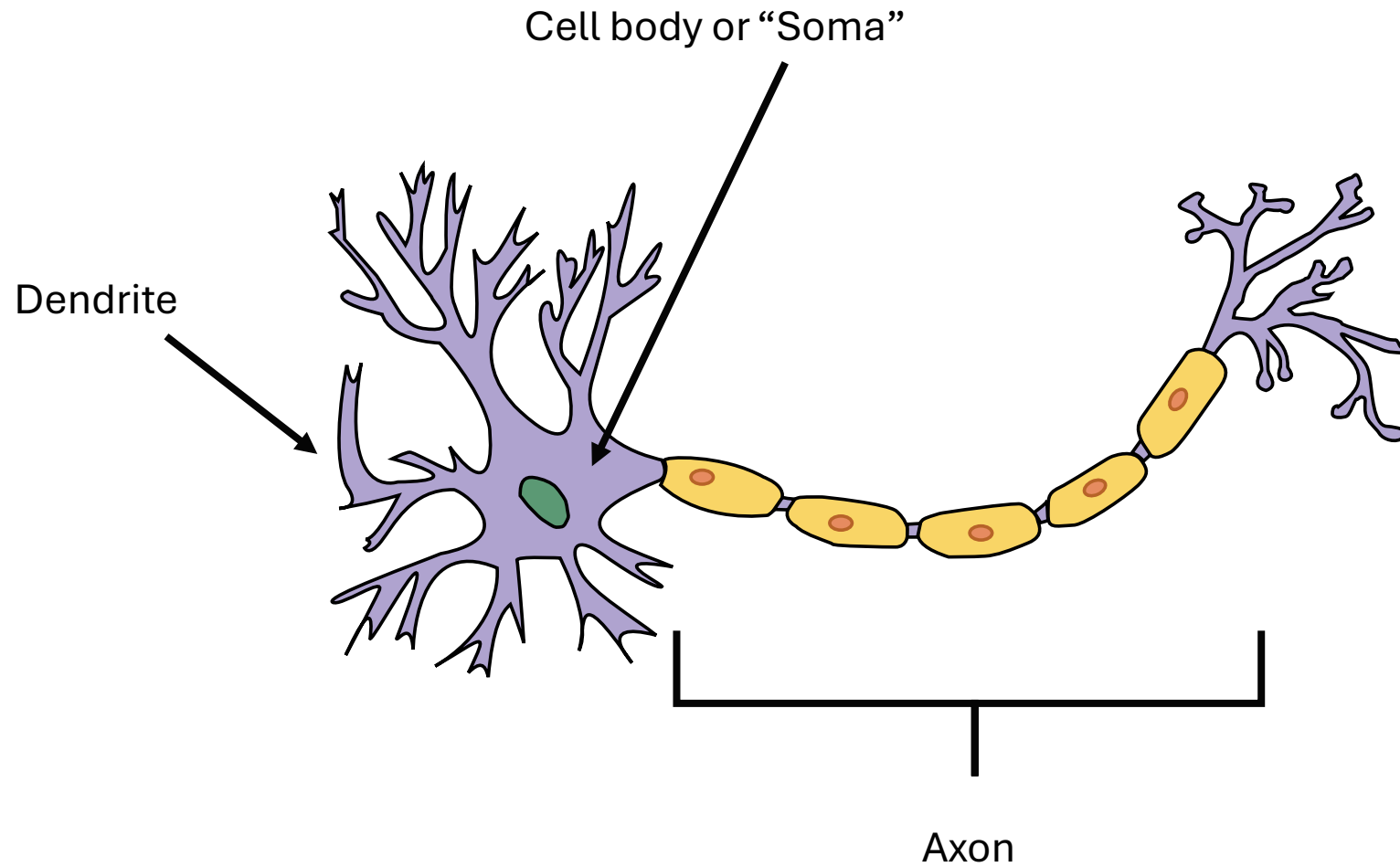
Avi Mohan

Neural Networks

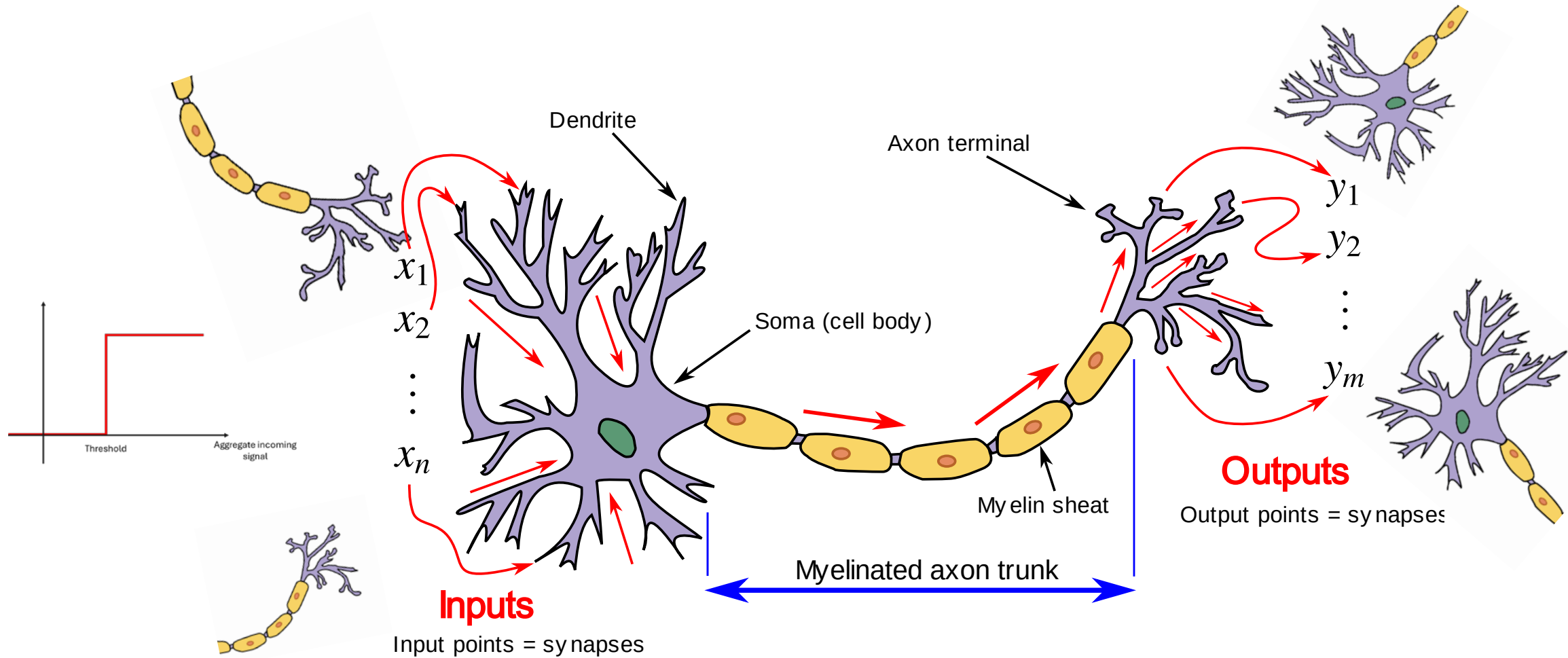
- Most early theory and architecture based on attempts to emulate the way the human brain functions
- Brain:
 - seat of intelligence
 - responsible for thoughts and actions
 - motor control
 - problem solving
 - navigation
 - speech
- can learn
- composed of nerve cells called “neurons”
 - vast network of interconnected neurons



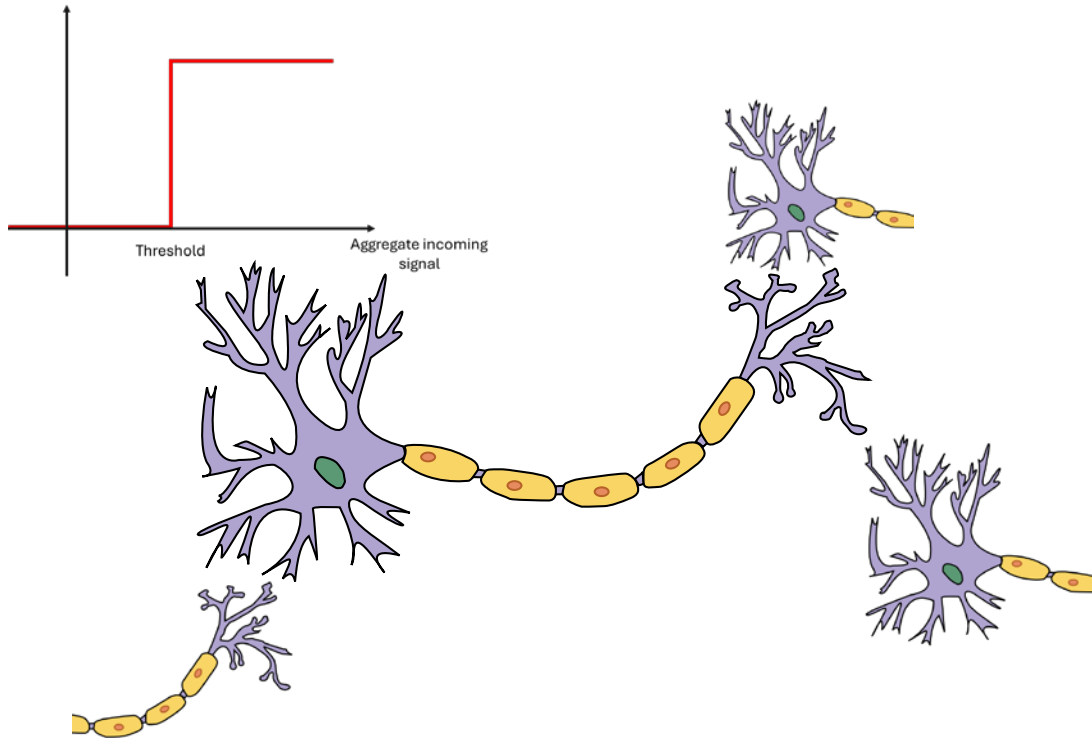
A Neuron



Neuron → Neural *Network*



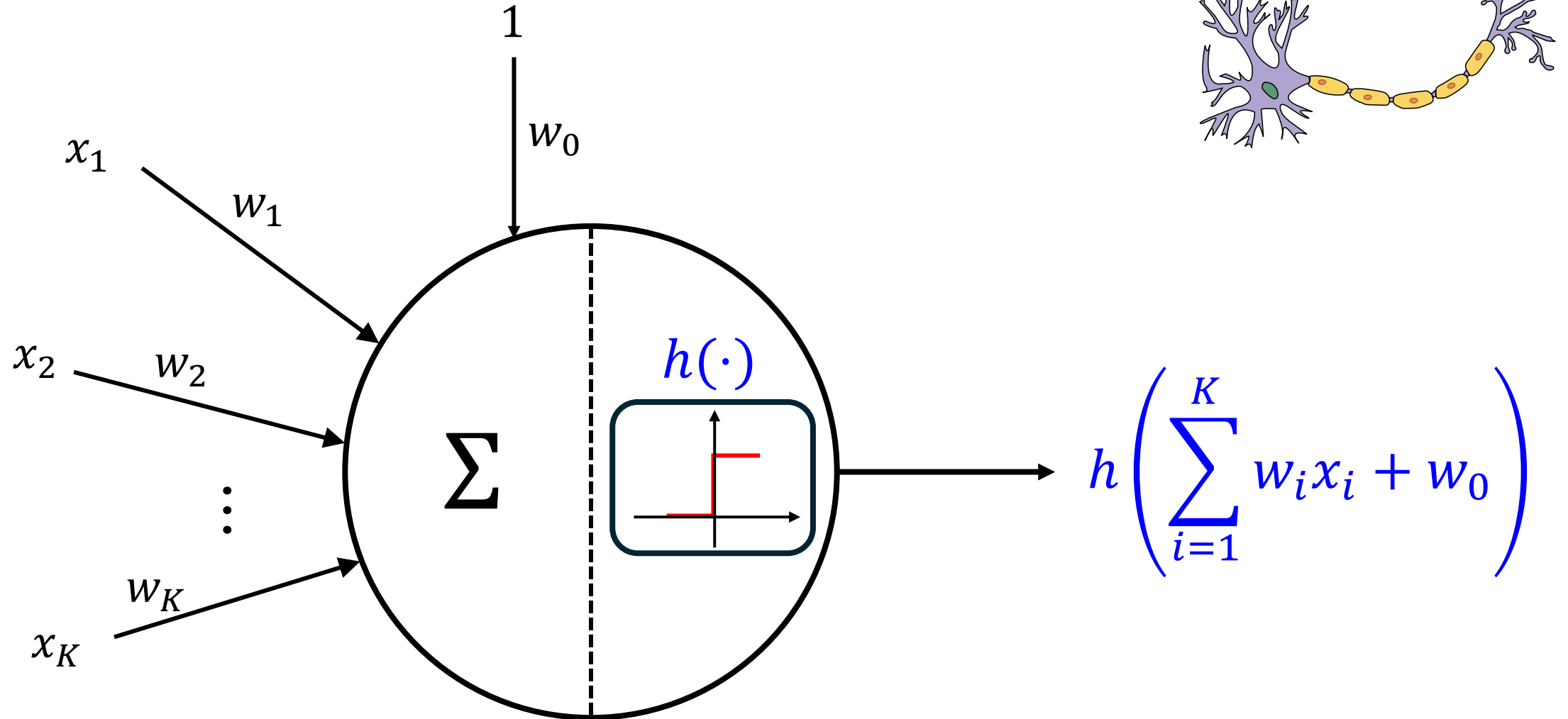
Neuron → Neural *Network*



- Two operations happen sequentially.
- **Weighted sum**
 - weights $\propto$ strength of connection
- **Nonlinearity**
 - Thresholding

The Artificial (MCP) Neuron

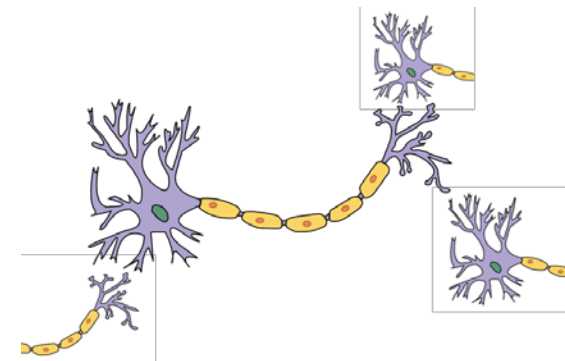
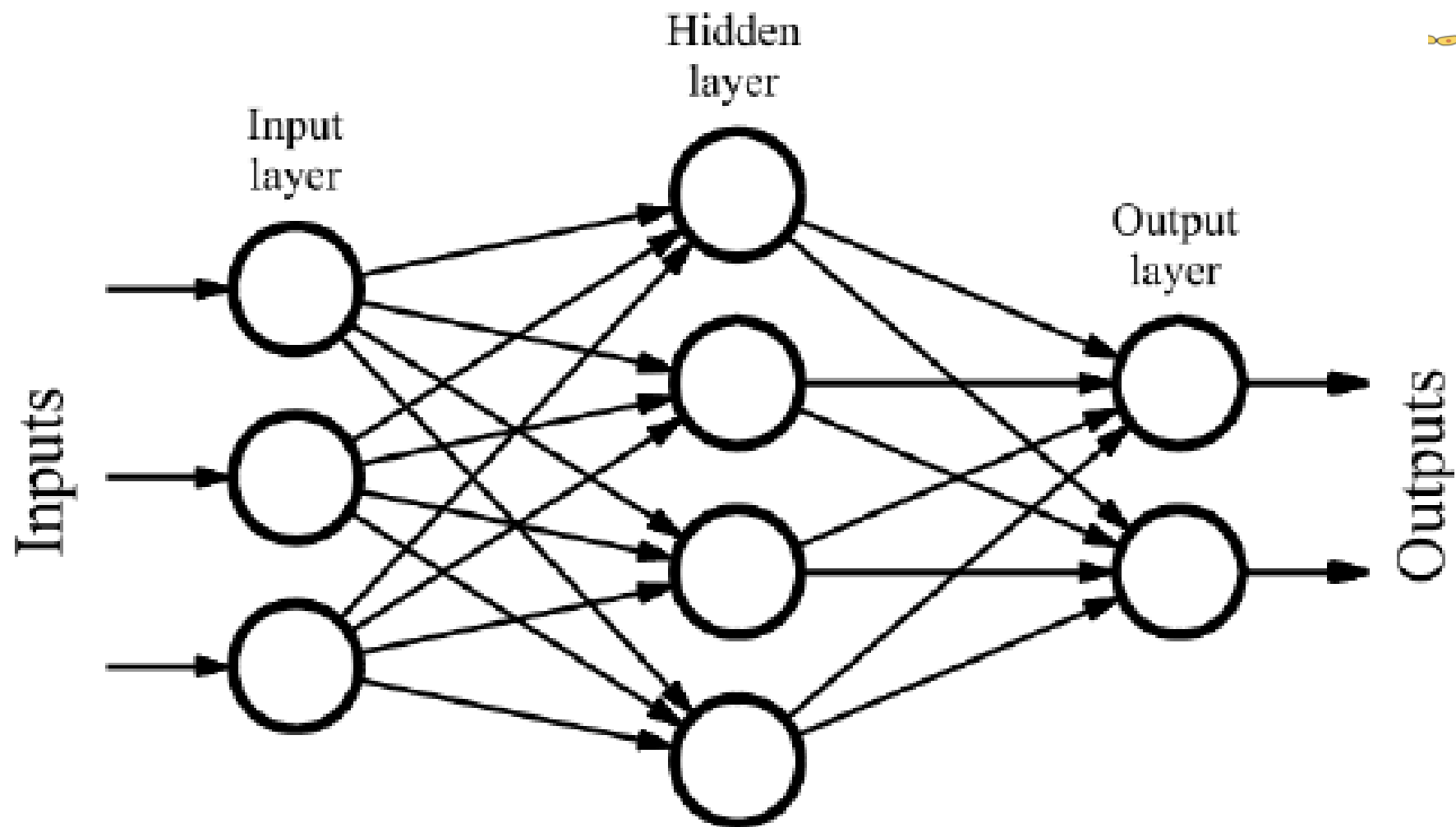
- Invented by Warren McCulloch and Walter Pitts (1943).



Single-Neuron Networks

- Classification and Regression using single neurons.

A Neural Network



Some Definitions

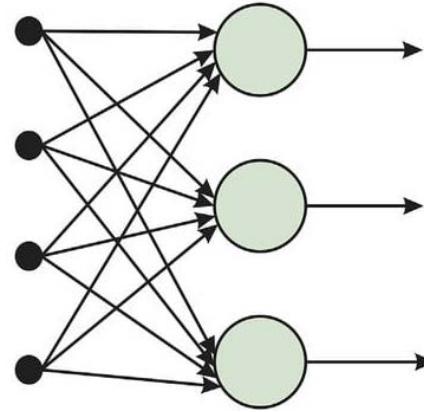
- Major terms:
 - Neuron
 - Layer
 - Input
 - Hidden
 - Output
 - Weights and Biases
 - Activation function

Some definitions (contd.)

- Architecture variations

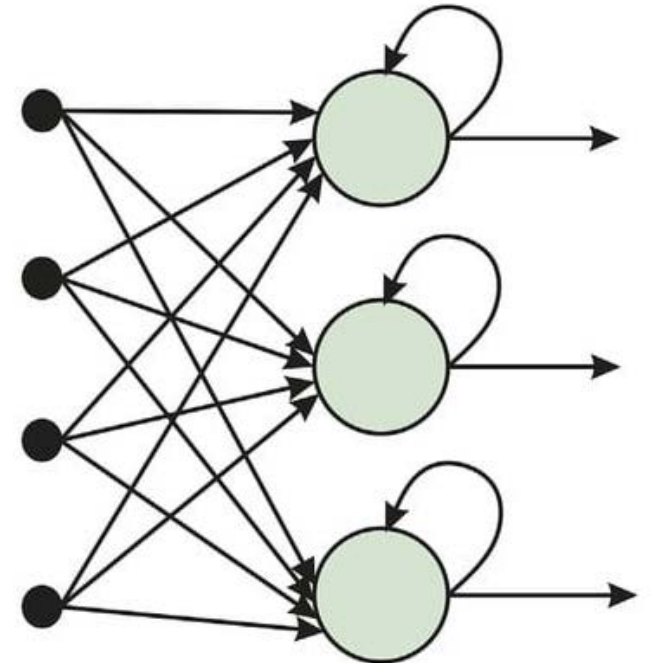
- **Feedforward Networks**

- Directed **Acyclic** Graph (DAG)
 - No internal state
 - Simply computes outputs from inputs

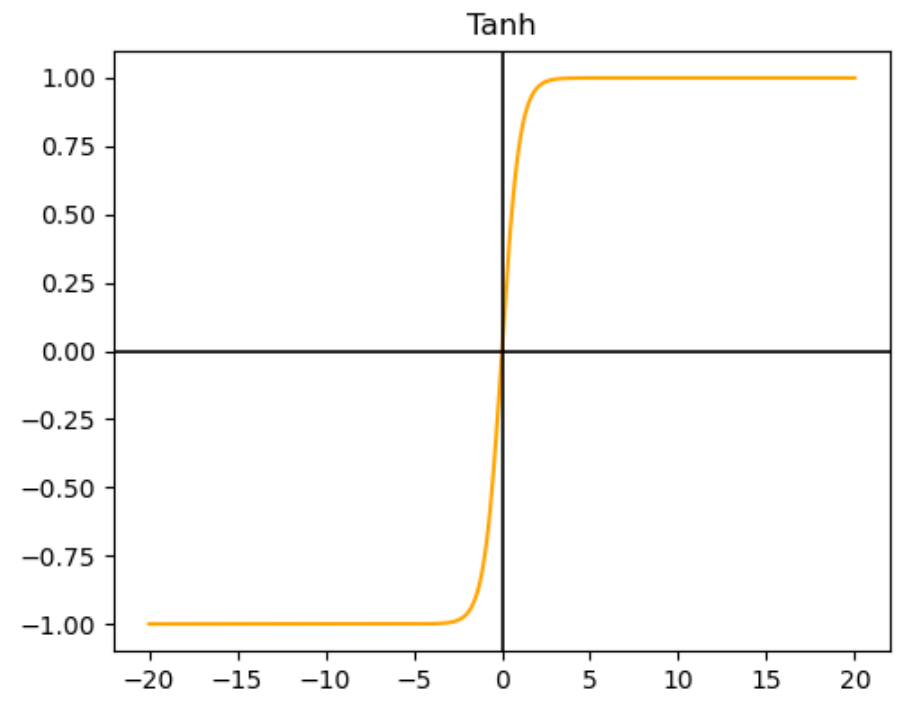
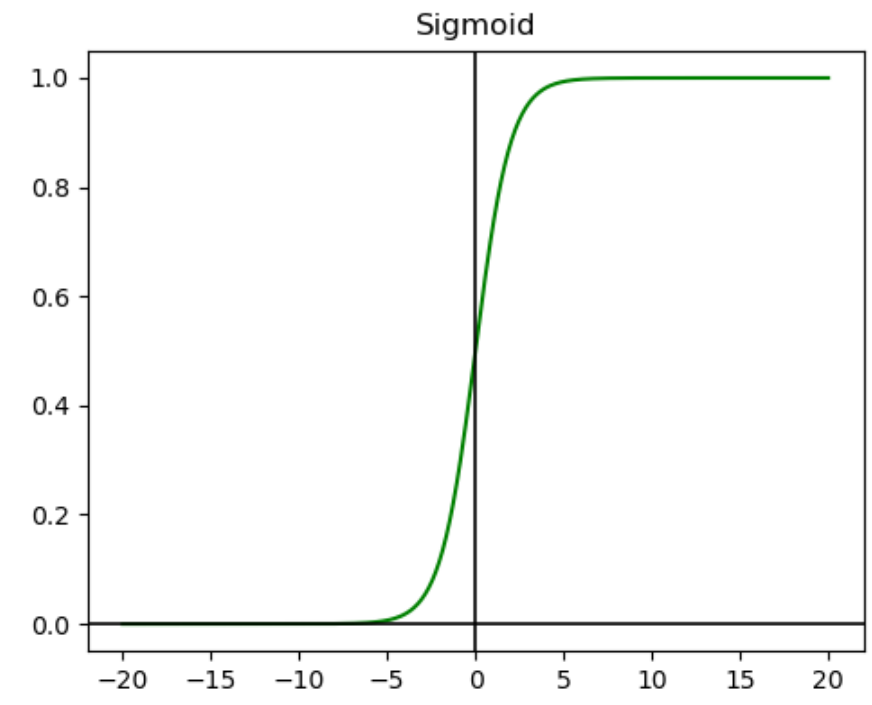
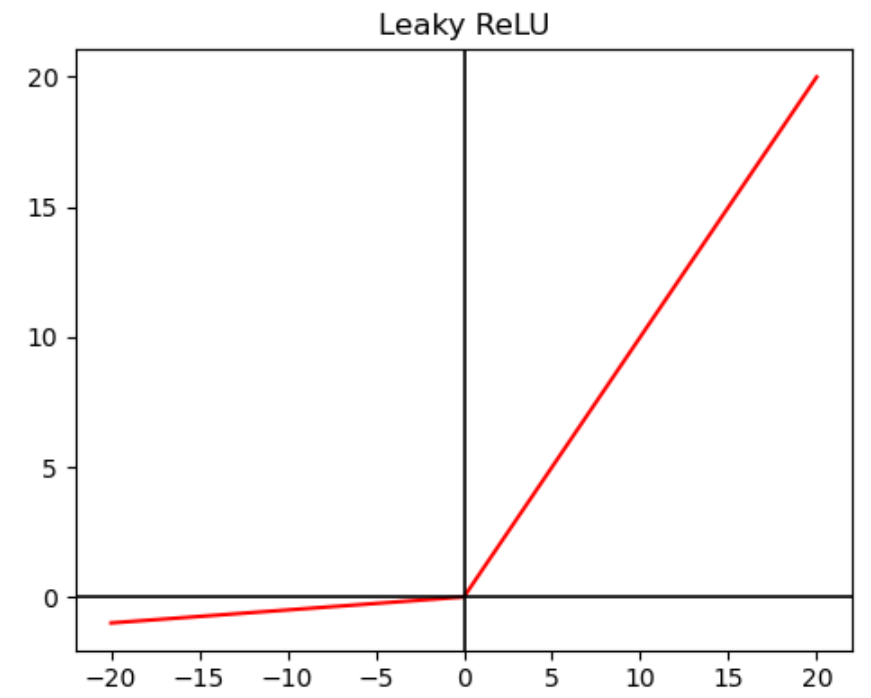
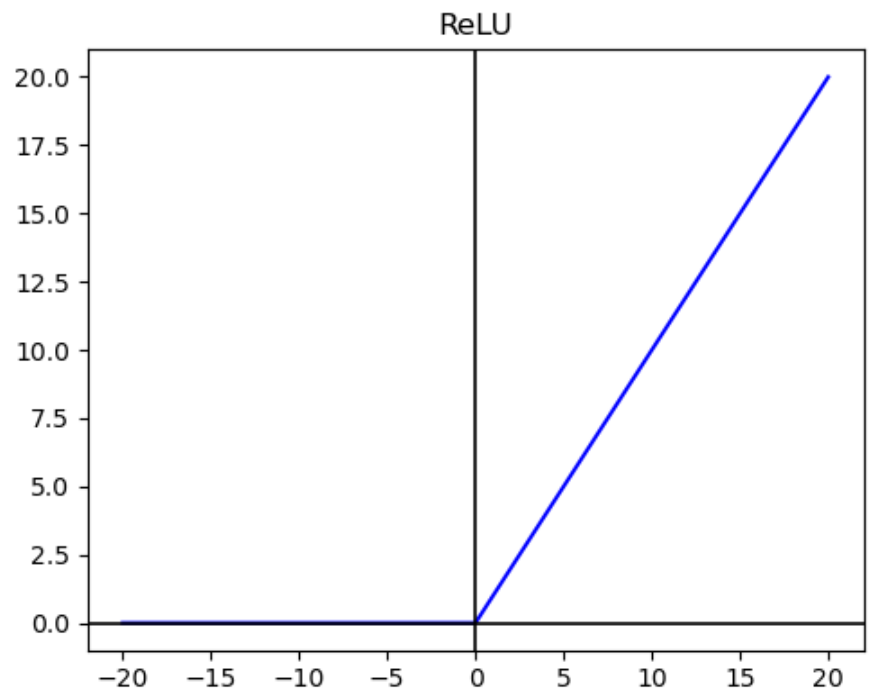


- **Recurrent Networks**

- Directed **Cyclic** Graph
 - Dynamical system with (evolving) internal states
 - Outputs are functions of both present and past inputs



Other activation functions



Other activation functions

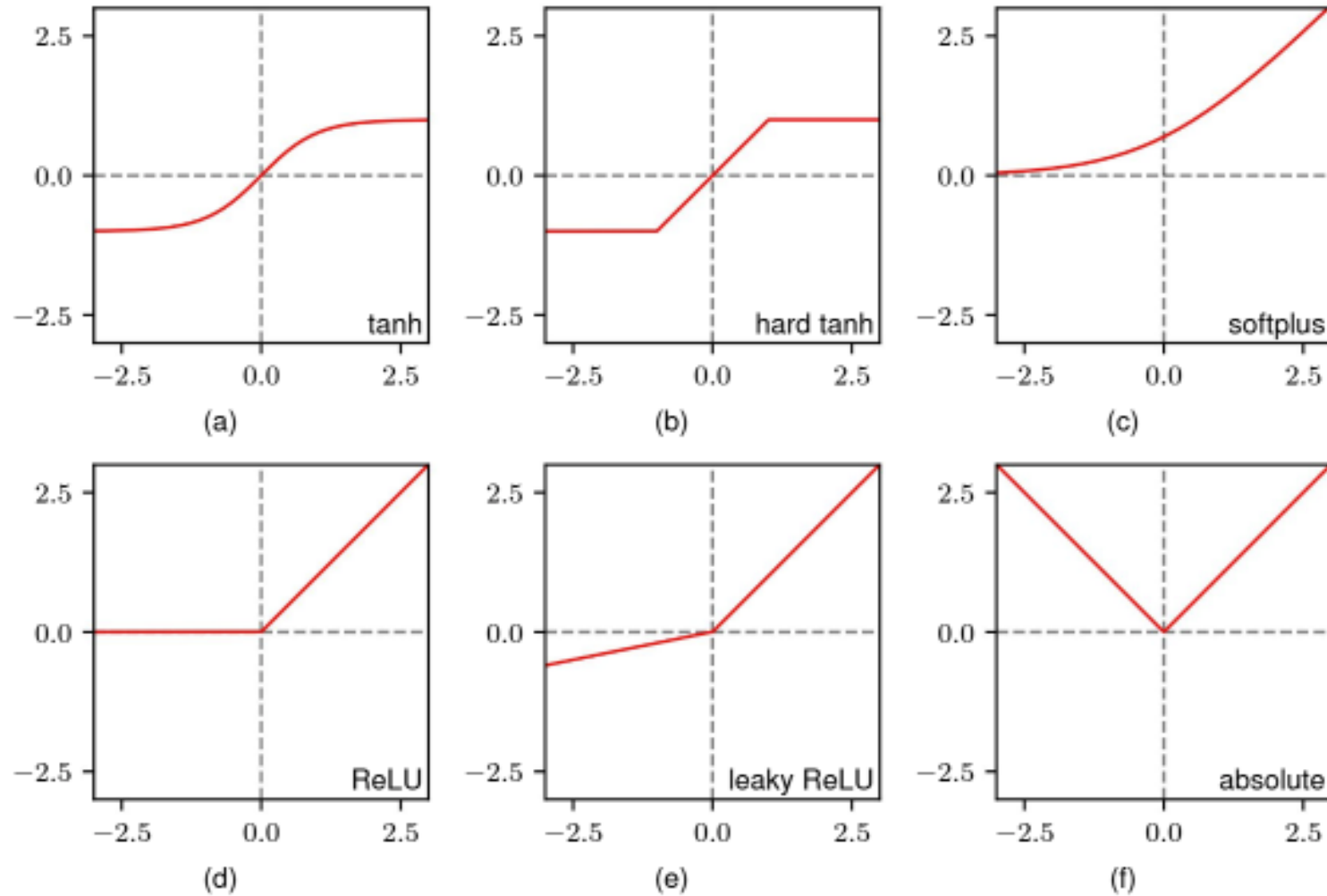


Figure 6.12 A variety of nonlinear activation functions.

Source: Sec. 6.2.3 of
Bishop's book.

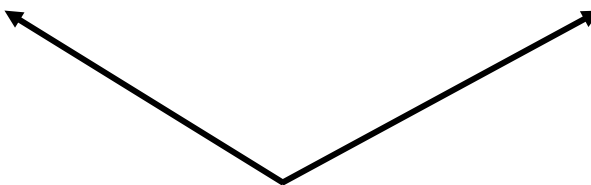
Logical Gates

- Two inputs: x_1, x_2
 - Assume both inputs are binary
 - $x_i \in \{0,1\}$
- One output: y
 - Also binary.
- Activation function: **threshold** (aka Heaviside)
 - $a = w_0 + w_1x_1 + w_2x_2$
 - $y = h(a)$

Logical Gates

- What is the **end-to-end function** being realized here?

$$y = \sigma(w_{31} \sigma(w_{11}x_1 + w_{12}x_2 + w_{01}) + w_{32} \sigma(w_{21}x_1 + w_{22}x_2 + w_{02}) + w_{03})$$



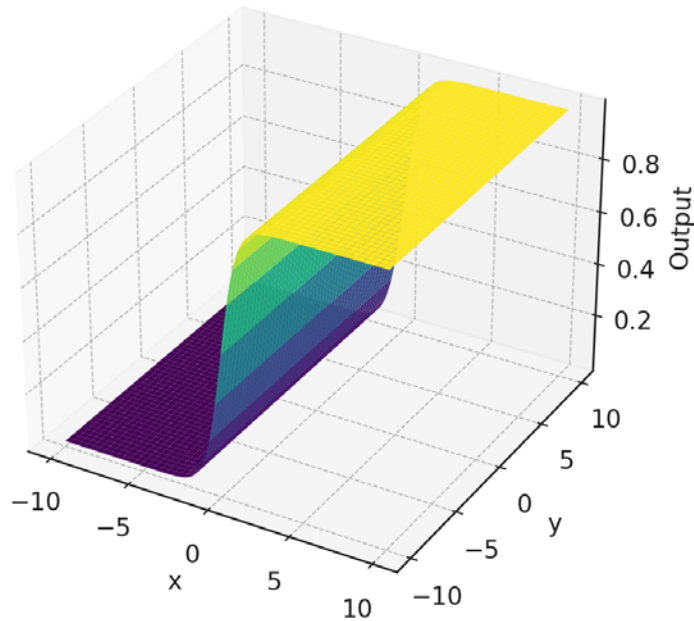
outputs of the two hidden neurons

Universal Approximation Theorem

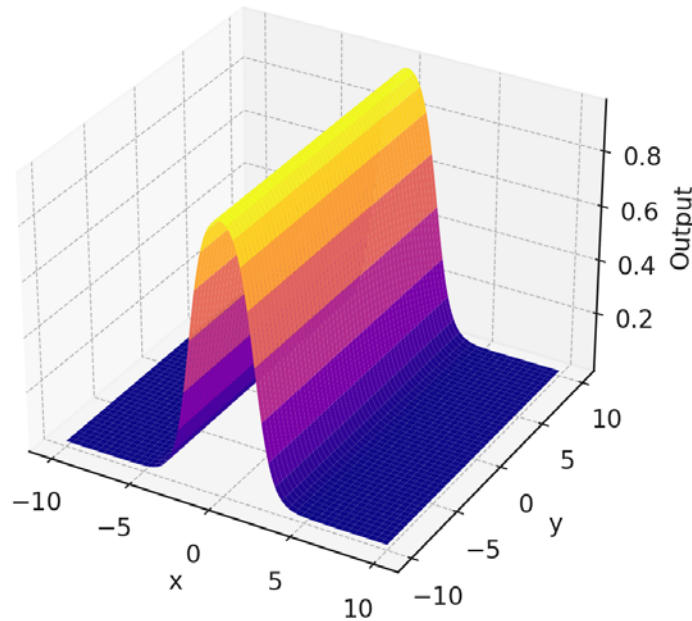
- Very informally, this result states that a FF network with a **single hidden layer** can approximate any function.
 - Caveat: the **number of neurons** in said hidden layer could be enormous.
 - **Continuity** is a requirement.
- Versions of this theorem for networks of **arbitrary width and depth** have been proven.
- Other versions for **various activation functions** also exist.

Universal Approximation Theorem

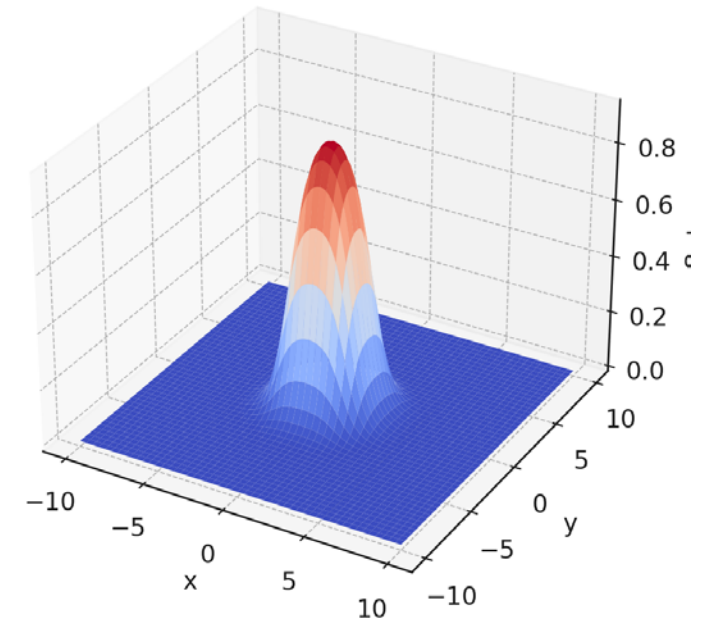
Cliff (One Sigmoid)



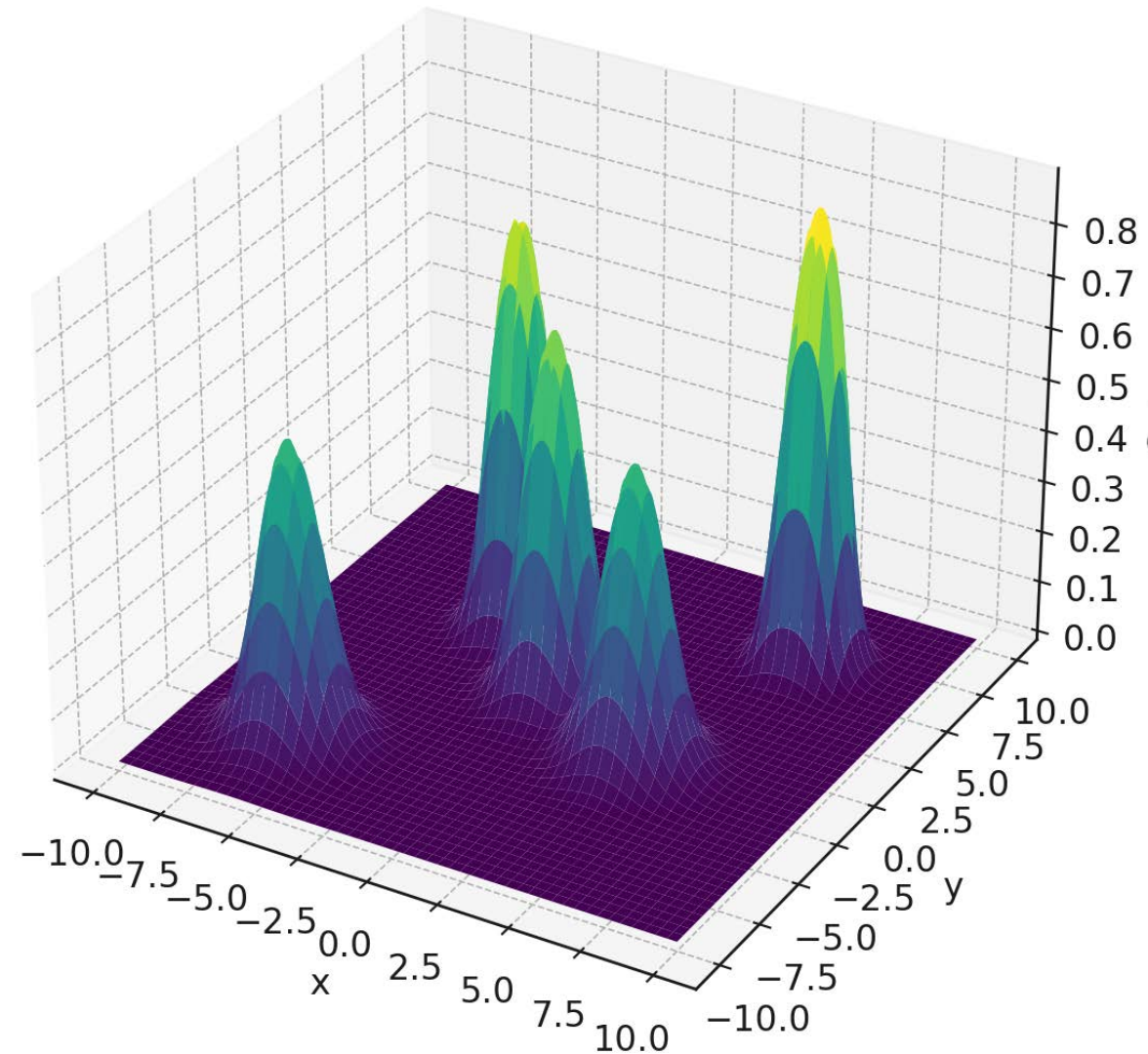
Ridge (Two Cliffs)



Bump (Two Ridges)



Universal Approximation Theorem



Universal Approximation Theorem

- **Sec. 6.4** in Ian Goodfellow's book has a succinct and clear discussion.
- Also, DeepMind's blog post on the universal approximation theorem: <https://www.deep-mind.org/2023/03/26/the-universal-approximation-theorem/>
 - this post also links to **3Blue1Brown's** excellent miniseries on deep learning (**a must watch!**)